

Foundation (Jalapeno)

Q1. What is the expansion of $(x + y)^2$?

- (A) $x^2 + 2xy + y^2$
 (B) $x^2 - 2xy + y^2$
 (C) $x^2 + y^2$
 (D) $2x^2 + 2xy + 2y^2$
 (E) $2x^2 - 2xy + 2y^2$

Q2. Which of the following is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
 (B) $x^2 - 4x - 4$
 (C) $x^2 + 4x - 4$
 (D) $x^2 - 4x + 4$
 (E) $x^2 + 2x + 1$

Q3. What is the factored form of $x^2 + 6x + 9$?

- (A) $(x + 3)^2$
 (B) $(x - 3)^2$
 (C) $(x + 3)(x - 3)$
 (D) $(x^2 + 3)$
 (E) $(x + 3)^3$

Q4. What is the expansion of $(x - y)^2$?

- (A) $x^2 - 2xy + y^2$
 (B) $x^2 + 2xy + y^2$
 (C) $x^2 - y^2$
 (D) $2x^2 - 2xy + 2y^2$
 (E) $2x^2 + 2xy + 2y^2$

Q5. Which of the following is NOT a perfect square trinomial?

- (A) $x^2 + 4x + 4$
 (B) $x^2 - 4x + 4$
 (C) $x^2 + 4x - 4$
 (D) $x^2 - 4x - 4$
 (E) $x^2 + 2x + 1$

Q6. What is the factored form of $x^2 - 8x + 16$?

- (A) $(x - 4)^2$
 (B) $(x + 4)^2$
 (C) $(x - 4)(x + 4)$
 (D) $(x^2 - 4)$
 (E) $(x - 4)^3$

Q7. What is the expansion of $(2x + 3y)^2$?

- (A) $4x^2 + 12xy + 9y^2$
 (B) $4x^2 - 12xy + 9y^2$
 (C) $4x^2 + 9y^2$
 (D) $12x^2 + 12xy + 12y^2$
 (E) $12x^2 - 12xy + 12y^2$

Q8. What is the expansion of $(3x - 2y)^2$?

- (A) $9x^2 - 12xy + 4y^2$
 (B) $9x^2 + 12xy + 4y^2$
 (C) $9x^2 - 4y^2$
 (D) $12x^2 - 12xy + 12y^2$
 (E) $12x^2 + 12xy + 12y^2$

Open Ended Questions (FRQ)**Q9.** Factor the perfect square trinomial $x^2 + 10x + 25$ and show your work.**Q10.** Expand the binomial $(x + 2y)^2$ and simplify the expression.

Chef's Tips

- Read the questions carefully
- Use the proper format for your answers
- Check your work before submitting